Calibrated probabilistic forecasting of football match outcomes using machine learning and advanced feature engineering

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***Abstract*—Football match prediction has been gaining increasing popularity as a classification problem in recent studies, yet the value of any predictive model lies in finding the right balance between correctly identifying the outcome and making sure that the predicted probabilities reliably reflect the actual likelihood of these outcomes. Rather than treating the task of predicting football match outcome purely as a classification problem, this paper discusses a machine learning framework for predicting football match results with a particular focus on both predictive accuracy and the quality of predicted probabilities, generating calibrated estimates for three outcomes: home win, draw, and away win. Historical data from La Liga matches were used to engineer a feature set capturing match context and team strength. Four models were trained and compared, including Logistic Regression, Random Forest, XGBoost, and AutoGluon, with AutoGluon achieving the highest classification accuracy of 64.3%. Models were evaluated progressing from standard measures such as accuracy and log-loss to calibration analysis using Expected Calibration Error, Brier Score, and Ranked Probability Score, benchmarked against bookmaker-implied probabilities with an ECE of 0.036. Temperature Scaling reduced XGBoost's ECE from 0.048 to 0.033, outperforming the bookmaker benchmark. A key finding is that classification accuracy and probability calibration are independent properties, with direct implications for model selection in any application where probability estimates drive financial or strategic decision-making.**

***Index Terms*—football match prediction, probability calibration, machine learning, probabilistic forecasting**

I. Introduction

Football match result prediction has become increasingly popular in the fields of sports analytics and machine learning over the past decade. Most of the existing solutions address this task as a supervised classification task: given a set of pre-match and in-play features, a model predicts one of two or three outcomes: a home win or not a home win, or, in the case of a three-way outcome, a home win, draw, or an away win. This framework has become a useful tool for benchmarking various algorithms performance, and leading-edge models often achieve accuracies in the range of 60 to 65 percent on major European league data [1] [2] [3].

However, framing football match prediction purely as a classification task often results in fundamental mismatch when the intended application uses probabilistic decision-making. In risk assessment, betting markets, or any other context where decisions are based not only on the predicted class label, but also magnitude of predicted probabilities, the reliability of these probabilities becomes one of the most important metrics of the model.

The distinction between predicting the right outcome and predicting a reliable probability is formally captured by the concept of calibration. A model is considered well-calibrated if its predicted probabilities correspond to the actual frequencies of observed outcome. Formally, among all events assigned predicted probability p, approximately a proportion p of those events should actually occur. Calibration is therefore not a property of the most likely predicted class, but a property of the entire distributional profile, and it cannot be inferred from the model accuracy alone.

Despite the importance, probability calibration remains underdeveloped in football prediction. Most of the existing studies pervasively evaluate models using accuracy, F1 score, or simulated betting profit. While these metrics measure whether the predicted outcome is correct, they do not explicitly measure the reliability and trustworthiness of the predicted probabilities. Most existing machine learning models produce probability estimates as an architectural consequence, yet this information is rarely subjected to calibration diagnosis. As a result, the question of whether the predicted probabilities of common football match predicting models can be trustworthy and serve as a basis for strategic decision-making remains largely unaddressed.

Addressing that gap directly, this paper's pivotal contribution consists of a systematic framework for evaluating the probabilistic robustness of machine learning models in football match prediction, combining calibration as a primary performance metric alongside classification performance. The framework is applied to Spanish La Liga match data covering ten seasons spanning from 2015/2016 to 2024/2025, using four models across a gradient of architectural complexity: Logistic Regression, Random Forest, XGBoost, and AutoGluon. Bookmaker-implied probabilities serve as a calibration

benchmark, providing a functional point of reference that reflects the market-implied consensus of the active forecasting markets.

In addressing this gap, this paper first compares four machine learning models in terms of both classification metrics and calibration diagnosis, including Expected Calibration Error (ECE), Maximum Calibration Error (MCE), Brier Score, and Ranked Probability Score. Secondly, three post-hoc calibration methods including Platt Scaling, Isotonic Regression, and Temperature Scaling are applied for the models that need calibration correction. Then, model-implied probabilities are benchmarked against bookmaker-implied probabilities to assess operational relevance. In the end, the paper provides evidence that model accuracy and model reliability are independent properties, and that the selected model should account for both dimensions depending on the intended application. To this end, this paper addresses the following research questions:

- Do standard machine learning models produce well-calibrated results for football match outcomes, and how do their produced probabilities compare to market-implied probabilities across three outcome classes?
- Can post-hoc calibration methods improve the probabilistic quality of poorly calibrated models without degrading classification accuracy?

II. Related work

A wide range of machine learning models have been applied to football match outcome prediction across major European Leagues. These models include Random Forest, Support Vector Machine (SVM), XGBoost and its expansions, Neural Networks, and Long Short-Term Memory architectures [1] [2] [3]. These predictions typically integrate historical match statistics, team ratings (FIFA, Elo), player attributes, and bookmaker odds, with state-of-the-art models achieving accuracies in the range of 63 to 65 percent on English Premier League and international match data. By addressing class distribution, balanced sampling techniques have been proven to extensively increase predictive performance, with binary classifications consistently outperforming multiclass (3-class) predictions in multiple experimental setups [3]. Apart from accuracy, some studies suggest that machine learning models can generate positive betting returns under simulation and in certain setups even outperform bookmaker prediction performance in terms of accuracy [4]. With the development of advanced team and player performance metrics such as expected goals (xG) and dynamic team ratings, the prediction accuracies have been consistently bolstered relative to traditional goal-based approaches [5]. However, these studies share a common limitation: model selection and evaluation are driven entirely by classification performance, without examining whether the predicted probabilities themselves are reliable. Rodrigues and Pinto [1] evaluate performance through accuracy and profit simulation only, Rahman [2] focuses on predictive accuracy without any probabilistic assessment, and Choi et al. [3] similarly rely on accuracy, F1 score, and AUC as primary criteria. As a result, whether the probability outputs of these models can be trusted for decision-making remains an open question.

Despite the expansive sense of this literature, evaluation of model performance remains primarily classification-focused. Model performance is typically evaluated through accuracy, F-1 score, area under the curve, simulated return on investment (ROI), and the probabilistic reliability of outputs is rarely examined.

An emerging but expanding area of work has shifted the attention from pure classification accuracy to the quality of probabilistic performance in football prediction. Calibration in different models has been shown to be dependent on the model itself: some models benefit from post-hoc calibration, others experience performance downgrade, and careful hyperparameter tuning to optimize log-loss can also yield well-calibrated results without additional calibration correction [6].

Wheatcroft and Sienkiewicz [6] evaluate seven machine learning models across 35,733 matches from 22 European leagues, finding that Platt Scaling improved calibration only for certain models such as Gaussian Naive Bayes and AdaBoost, and actually reduced performance for others, establishing that post-hoc calibration should not be automatically applied but must be tested on a per-model basis. Beyond calibration correction, the study introduces a conceptually important shift: rather than optimizing hyperparameters for log-loss, models are tuned directly for betting profit using value betting and Kelly strategies. While forecast skill and profit are related, the relationship is shown to be imperfect, suggesting that model selection should account for the intended decision context rather than a single universal metric. Neural Networks achieved the best performance in both log-loss and betting profitability, further supporting the argument that architectural complexity and proper score optimization interact in non-trivial ways. The choice of evaluation criterion is itself consequential: the Ranked Probability Score, widely used in sports analytics, has been shown to be less efficient than the logarithmic score for the identification of robust forecasting frameworks, and its adoption of outcome proximity as a reward metric lacks theoretical justification [7].

Wheatcroft [7] compares three proper scoring rules: the Ignorance score (logarithmic score), the Brier Score, and the Ranked Probability Score, across two simulation experiments involving over 39,000 football matches. The central argument is that sensitivity to distance, the defining property of the RPS, does not align with the fundamental purpose of scoring rules, because once an outcome is observed, no information is revealed about the probabilities of unrealised outcomes. The Ignorance score, being local, consistently identifies superior forecasting systems with fewer observations than both the Brier Score and the RPS, particularly when differences between candidate forecasts are subtle. From an information-theoretic perspective, the Ignorance score is closely related to log-likelihood, inheriting desirable inferential properties that are absent in the RPS. This has direct implications for model comparison in sports analytics, where near-optimal forecasting systems are common and evaluation efficiency is critical. Score

decomposition methods show that low accuracy scores in football prediction originate from limited resolution rather than miscalibration, especially for draw outcomes [11].

Foulley [11] extends the evaluation of probabilistic football forecasts beyond mean scoring rules by applying Murphy's calibration-refinement decomposition to UEFA Champions League forecasts generated from a Poisson regression model with ELO-based covariates, compared against bookmaker-implied probabilities. Results show that both approaches are generally well-calibrated, but bookmakers exhibit slightly stronger resolution and discrimination, especially for home and away wins, while draw forecasts consistently display the weakest resolution across all evaluated approaches. A key finding is that most forecast deficiencies originate not from miscalibration but from limited resolution, meaning that models assign probabilities that are reliable on average but insufficiently sharp to discriminate between outcomes. This methodological contribution shifts attention from simple accuracy comparisons toward understanding why forecasts succeed or fail, a perspective directly relevant to the calibration framework developed in this paper. Bayesian approaches, in contrast, have yielded calibration outcomes that are competitive relative to traditional benchmarks [8].

Diniz et al. [8] develop two Bayesian multinomial-Dirichlet models for predicting match outcomes in the Brazilian first division, operating directly on categorical outcome counts rather than detailed covariates. The models exploit Dirichlet-multinomial conjugacy to derive closed-form posterior predictive probabilities, and combine home and away information through linear opinion pooling. Empirical evaluation across 1,710 matches shows that these models achieve calibration performance comparable to or better than Bradley-Terry and Poisson-based benchmarks under proper scoring rules including the Brier Score and logarithmic score. A key contribution is the emphasis on probabilistic evaluation rather than classification accuracy, with the authors explicitly assessing calibration and concluding that some competing models overestimate probabilities for frequent outcomes. However, the framework assumes independence between matches and relies solely on aggregate outcome counts, ignoring dynamic team strength evolution, player-level covariates, and contextual variables, limiting its applicability to more complex prediction settings.

From a wider machine learning perspective, Guo et al. [9] show that modern neural networks tend to be overconfident even with high classification accuracy performance. Their study shows that Temperature Scaling, a simple one-parameter post-hoc correction, consistently reduces the Expected Calibration Error without harming the overall classification performance.

This overconfidence arises from increased model depth, reduced regularization, and the use of Batch Normalization, and is observed even when classification accuracy continues to improve, demonstrating that accuracy and calibration can diverge structurally rather than as an isolated empirical result. Among all evaluated post-hoc methods including histogram binning, isotonic regression, Platt scaling, vector scaling, and matrix scaling, Temperature Scaling provided the most effective and computationally efficient calibration across tasks, adjusting softmax outputs without changing classification decisions. This finding is directly linked to ensemble models used in football prediction, which show similar probability performance patterns. Synthesizing 219 studies, a large-scale systematic review reveals that in sports analytics and especially betting research the preference is still given to classification accuracy, while the call for stronger probabilistic standards and strategic decision-oriented frameworks is gaining significant momentum [10].

Galekwa et al. [10] identify recurring challenges across this body of work, including data quality limitations, lack of standardized evaluation metrics, overfitting, and weak out-of-sample validation, and argue for a shift toward decision-oriented frameworks using portfolio-based betting strategies to better balance return and risk. Importantly, the review highlights that many studies still prioritize classification accuracy despite evidence that accuracy alone does not ensure betting profitability, a finding that directly motivates the calibration-first framework developed in this paper.

Existing calibration research in football forecasting tends to focus on specific models, binary formulations, or single-competition setups. Systematic comparison of post-hoc correction methods, comprehensive multidimensional calibration profile, and direct benchmarking against market-implied probabilities remain rare. This paper addresses that gap directly by treating calibration as a primary performance criterion, applied consistently across four models on a decade of Spanish La Liga historical data [13].

Despite its contributions and the depth of surveyed literature, this study has several limitations that distinguish it from other work. The developed framework is restricted to a single league, Spanish La Liga, and a single season of 380 matches, which limits the generalization of the findings across different contexts. Furthermore, the feature set is limited to pre-match features and does not incorporate in-play data, player-level statistics, and the study does not employ LSTM architecture or neural networks. These aspects are reserved for future work.

III. Methods

A. Dataset and experimental setup

The dataset includes historical data from ten seasons of Spanish La Liga, spanning from season 2015/2016 to 2024/2025, yielding a dataset of 3,699 matches in total [13]. A strict temporal split was used for training and testing to reflect realistic deployment conditions and prevent data leakage: all models were trained chronologically on seasons up to and including 2023/2024, corresponding to 3,319 matches, and tested on the held-out 2024/2025 season, including 380 matches. The target variable was set to full-time result, represented as one of three possible discrete outcomes: home win, draw, and away win, with class frequencies of 45.5%, 26% and 28.5% respectively in the training set, highlighting the well-documented home advantage effect in football.

The overall implementation follows a sequential pipeline. Raw match data is first collected and preprocessed, after which features are constructed through domain-driven feature engineering. Statistical selection methods, including ANOVA F-tests and VIF analysis, are then applied to retain only significant and independent signals. The selected feature set is used to train four machine learning models on historical seasons, and classification performance is evaluated on the held-out test season using accuracy, F1 score, and log-loss. For models identified as poorly calibrated, post-hoc calibration methods are applied, and calibration quality is subsequently evaluated using Expected Calibration Error, Brier Score, and Ranked Probability Score, benchmarked against bookmaker-implied probabilities.

B. Feature engineering

The original feature set was built through exploratory data analysis combined with domain knowledge, leveraging theoretically grounded indicators from analyzed literature. After statistical selection, fifteen features across five groups were chosen as a main feature set. Among these, bookmaker-implied probabilities for three outcomes were included, converted from raw odds after removing the over-round margin. The team strength group captured the Elo rating difference between two teams, providing a dynamic metric of team quality updated after every match. The form group includes rolling differentials in points earned in the last five matches, where a win is awarded three points, a draw one point, and a loss no points, following standard football competition rules, defensive and attack performance, and goals scored. The tactical performance group included expected point difference, passes per defensive action, deep completions, and shots on target difference. Finally, the underlying group captured longer-term offensive and defensive tendencies, including season-average expected goals for home and away teams. The final feature set was formalized using two statistical methods. ANOVA F-tests were applied to remove statistically insignificant features, keeping only features that have statistically significant correlation with the target variable. VIF analysis was then conducted to eliminate multicollinear features, ensuring that only independent signals are included in the final set. In addition to the three-class formulation of the task, a binary reformulation with possible outcome Home win versus Not home win is introduced as a complementary experiment. Given the well-documented difficulty of draw class prediction, the three-class method was simplified to binary class to assess whether the feature set retains strong discriminative power when the class imbalance is reduced. The binary reformulation is evaluated using F1-score, ROC-AUC and accuracy.

C. Models

All models were implemented in Python using the scikit-learn library for Logistic Regression, Random Forest, and calibration methods, the XGBoost library for gradient boosting, and the AutoGluon framework for automated machine learning.

Four models were evaluated, representing a diverse range of architectural complexity. As a principled baseline, Logistic Regression with L2 regularization was used, as it directly optimizes the log-loss score during training, which naturally yields well-calibrated results [6]. Random Forest was tuned through minimum leaf size to prevent probability compression that usually arises from vote-fraction averaging [12], with a final configuration of maximum depth 10 and minimum leaf of 20. XGBoost was tuned via systematic grid search to minimize multi-class log-loss. XGBoost was selected over other gradient boosting alternatives such as LightGBM due to its well-established performance on tabular classification tasks and its native support for multiclass log-loss optimization, making it a natural fit for the calibration-focused experimental setup of this paper. AutoGluon, an automated machine learning framework, constructs a weighted ensemble across multiple base learners automatically; the final configuration used the best-quality preset with a budget of 1,800 seconds. AutoGluon was included to provide an upper bound on ensemble performance without manual tuning, serving as a reference point for what automated model selection and stacking can achieve relative to individually tuned models.

D. Calibration methods

After the evaluation of models, three post-hoc calibration techniques were applied for Random Forest and XGBoost, each fitted using five-fold cross-validation to prevent leakage. Platt Scaling fits a logistic regression model on raw model scores to adjust output probabilities [12]. AutoGluon was excluded from external post-hoc calibration as it natively incorporates Temperature Scaling through its built-in calibration parameter, which automatically applies probability calibration on the validation set during the fitting process, making additional external correction redundant within this experimental setup. Isotonic Regression applies a non-parametric monotonic transformation to predicted probabilities and is known to require large datasets to perform reliably [12]. Despite this limitation, Isotonic Regression was included to provide a comprehensive comparison of post-hoc correction methods and to empirically confirm whether its known unreliability on small datasets is reproduced in this experimental setup. Temperature Scaling, introduced by Guo et al. [9], divides model logits by a single learned scalar parameter T before applying the softmax function:

$$\hat{p}_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \tag{1}$$

where z_i are the raw logits for class i and T is the temperature parameter optimized on a held-out validation set. A temperature below 1 sharpens the probability distribution and corrects systematic under-confidence, while a temperature greater than 1 softens it, reducing overconfidence [9]. Temperature Scaling is considered well-suited for calibration correction for applications where classification performance must be preserved [9].

E. Model evaluation

Models were compared across two dimensions. Classification performance was assessed based on accuracy, per-class F-1 score, and ROC-AUC for Home win/No Home win binary reformulation of the task. For probabilistic reliability, five calibration metrics were computed. First, Expected Calibration Error (ECE) approximates the average difference between the predicted probability and observed outcome frequency across partitioned confidence intervals, where a smaller gap indicates better calibration and zero represents a perfect match between predicted and observed probabilities [9]. Maximum Calibration Error (MCE) captures the worst-case gap across all probability bins, providing an upper bound on miscalibration. The Brier Score measures the mean squared difference between predicted probabilities and binary outcome indicators, penalizing both miscalibration and poor sharpness [7]. The Ranked Probability Score extends this to ordered multiclass outcomes by penalizing probability mass assigned too far from the actual outcome [8]. Reliability diagrams provide a qualitative counterpart to numerical metrics by plotting predicted probabilities against observed frequencies graphically. Bookmaker-implied probabilities serve as the calibration benchmark throughout all experiments, with an observed ECE of 0.036.

IV. Results

All models were tested on a held-out La Liga season consisting of 380 matches. Table I summarizes the classification results of multi-class models alongside the betting market as a baseline. All the models consistently outperformed the bookmaker baseline, showing that the engineered feature set captures meaningful signals beyond what is already covered in the bookmaker odds. Among all the models, AutoGluon achieved the highest overall accuracy and the best log-loss, followed by Random Forest, Logistic Regression, and XGBoost in descending order.

Logistic Regression's strong log-loss performance in comparison with other models suggests that the nature of the feature set is largely linear in essence. This observation is further supported by the fact that XGBoost underperformed both Logistic Regression and Random Forest across all classification metrics, despite its architectural complexity.

Predicting draw was consistently the most challenging for all the models, as shown in the F1 Draw and Draw Recall columns in Table I. This is consistent with the wider football analytics literature, where draw outcomes have lower base rate and higher outcome uncertainty [7] [9]. Home win prediction was the strongest across all models, showcasing the class frequency imbalance as described in the previous section.

A. Binary classification

Given the persistent difficulties to predict draw results across all models, a binary reformulation of the task was introduced as a complementary approach. By changing the multiclass setting to Home Win vs. Not Home Win reformulation, the three-class imbalance was reduced and the model can focus on the most frequent outcome class. Under this reformulation, all four models consistently achieved substantially higher results, as reported in Table II. The uniformly high performance across all models confirms that the feature set is specifically informative for predicting home win, and suggests that the significant portion of prediction errors are closely tied to the hard prediction of the draw class rather than the limitation of the feature set or model architecture.

TABLE I
MULTI-CLASS CLASSIFICATION RESULTS

Model	Acc.	LL	F1 H	F1 D	F1 A	DR
AutoGluon	0.643	0.831	0.746	0.451	0.657	0.469
Rand. Forest	0.637	0.848	0.753	0.319	0.640	0.227
Log. Reg.	0.629	0.845	0.759	0.256	0.636	0.175
XGBoost	0.616	0.857	0.738	0.197	0.654	0.134
Betting BL	0.544	-	-	-	-	-

Acc.=Accuracy, LL=Log Loss, H=Home, D=Draw, A=Away, DR=Draw Recall, BL=Baseline

TABLE II
BINARY CLASSIFICATION RESULTS (HOME WIN VS. NOT HOME WIN)

Model	Acc.	AUC	F1 HW	F1 NH	vs. BM
Rand. Forest	0.753	0.811	0.78	0.72	+13.8 pp
Log. Reg.	0.747	0.817	0.78	0.71	+13.2 pp
XGBoost	0.743	0.808	0.77	0.70	+12.8 pp
Bookmaker BL	0.615	-	-	-	Reference

Acc.=Accuracy, HW=Home Win, NH=Not Home, BM=Bookmaker, BL=Baseline

B. Calibration analysis

Table III summarizes calibration results before and after post-hoc correction for all models, benchmarked against bookmaker-implied probabilities, with an ECE of 0.036. Table IV provides a further breakdown of per-class calibration performance before and after Temperature Scaling.

Logistic Regression achieved the best natural calibration of all models before any correction, with an ECE of 0.0351, marginally better than the bookmaker benchmark. This outcome is consistent with the known property of logistic regression that directly optimizing log-loss during training naturally encourages well-calibrated probability outputs [6] [10]. No post-hoc correction was required. Per-class analysis in Table IV confirms that draw predictions achieved the best calibration of the three outcome classes with an ECE of 0.0157, while away win predictions showed the largest calibration gap before correction, a gap of 0.0549.

Random Forest showed the worst natural calibration results, notably worse than the bookmaker, despite outperforming it on all the classification metrics as shown in Table I. The model had the second highest accuracy at 0.637, yet is the worst naturally calibrated among all models. This is confirmed by the fact that classification accuracy and probability reliability are independent, and can diverge substantially in certain setups. Confidence quintile analysis confirmed that Random Forest systematically underestimated its own probabilities in every

confidence bin, which is a known consequence of vote-fraction probability compression in ensemble models [12]. Among three post-hoc methods assessed, Temperature Scaling yielded the best results, reducing ECE from 0.0453 to 0.0362, achieving the benchmark while also improving log-loss, Ranked Probability Score, and Brier Score all together. Platt Scaling, in contrast, provided only partial improvement, and Isotonic Regression performed poorly due to the limited size of the test data of 380 matches, consistent with its known requirement for large calibration datasets [12].

As shown in Table III, XGBoost showed the largest natural calibration error among all models, despite being tuned to minimize log-loss during hyperparameter search. This suggests that log-loss optimization alone does not guarantee well-calibrated outputs for gradient boosting models, unlike for Logistic Regression, which is naturally well-calibrated. Confidence quintile analysis showed systematic underconfidence across all portions, with high-confidence predictions having the largest gaps. The chosen Temperature Scaling parameter dropped below one, confirming that sharpening was required rather than softening. After calibration correction, XGBoost achieved the best post-calibration ECE of all models, reducing ECE from 0.0480 to 0.0327, demonstrating that a model built on bookmaker-derived features can be successfully re-calibrated to produce reliable probability estimates, despite having the second lowest classification accuracy as shown in Table I. The largest improvement for a specific class was observed for away team win predictions. However, the improvement was not uniform, draw calibration increased from 0.0222 to 0.0257, suggesting that uniform temperature parameters may not optimally correct calibration across all outcome classes.

In general, prior to any calibration correction, only Logistic Regression achieved the benchmark level calibration, due to its nature. All three models matched or exceeded the market benchmark after applying Temperature Scaling, showing that achieving well-calibrated results is feasible with standard machine learning models. Temperature Scaling was the most effective post-hoc calibration method, systematically improving the probability quality, while preserving the classification accuracy.

TABLE III
CALIBRATION RESULTS BEFORE AND AFTER POST-HOC CORRECTION

Model	ECE B	ECE A	Method	vs. BM	Verdict
Log. Reg.	0.0351	0.0351	None	Better	Good
Rand. Forest	0.0453	0.0362	T.S. (0.894)	Matches	Good
XGBoost	0.0480	0.0327	T.S. (0.748)	Better	Good
Bookmaker	0.0361	-	-	Ref.	Ref.

ECE B=Before, ECE A=After, BM=Bookmaker, T.S.=Temperature Scaling

TABLE IV
PER-CLASS ECE BEFORE AND AFTER TEMPERATURE SCALING

Model	Class	ECE Before	ECE After
Log. Reg.	Home Win	0.0347	0.0347
	Draw	0.0157	0.0157
	Away Win	0.0549	0.0549
Rand. Forest	Home Win	0.0558	0.0367
	Draw	0.0318	0.0299
	Away Win	0.0483	0.0421
XGBoost	Home Win	0.0597	0.0453
	Draw	0.0222	0.0257
	Away Win	0.0621	0.0272

V. ANALYSIS AND VALIDATION

A. Key findings

These results reveal three main findings. First, log-loss optimization during training is a meaningful step for calibration quality. As confirmed by the results for Logistic Regression, natural minimization of log-loss requires no post-hoc correction and produces the best probability reliability, consistent with the results of Wheatcroft and Sienkiewicz [6], who show that careful optimization of a proper score during initial training can fully substitute the calibration correction process.

Secondly, Temperature Scaling is the most recommended post-hoc calibration method for ensemble models in this particular experimental setup. It consistently outperformed other post-hoc calibration methods such as Platt Scaling and Isotonic Regression while preserving classification performance. Its single-parameter simplicity and computational efficiency make it well-suited for small-to-medium datasets in sports evaluation, and Isotonic Regression, in contrast, was shown to be unreliable [12].

Lastly, and most importantly, classification accuracy and calibration quality are mutually distinct characteristics of a machine learning model, as clearly illustrated in Tables I, III, and IV. Models with lower classification accuracy achieved the best post-hoc calibration ECE. This dissociation confirms that model selection must consider both dimensions separately depending on the intended application. For strategic decision-making applications where probability estimates drive financial decisions, calibration quality and probability reliability should be weighted at least as heavily as classification accuracy [6] [11].

B. Validation

The results of this paper are validated through three complementary approaches. First, a strict temporal split was applied throughout all experiments, ensuring that no future data was used during training and that the evaluation reflects realistic deployment conditions. Second, bookmaker-implied probabilities, with an observed ECE of 0.036, serve as a functional calibration benchmark, representing the market-implied consensus of professional forecasting markets and providing a meaningful reference point for assessing the probabilistic quality of model outputs. Third, five-fold cross-validation

was applied during post-hoc calibration fitting to prevent leakage between calibration and evaluation sets. Together, these validation approaches ensure that the reported results are not artifacts of data leakage or overfitting, and that the calibration improvements observed after Temperature Scaling reflect genuine improvements in probabilistic reliability.

C. Comparison with existing work

The classification results obtained in this paper are consistent with the wider football analytics literature, where leading-edge models achieve accuracies in the range of 63 to 65 percent on major European league data [1] [2] [3]. AutoGluon's best accuracy of 64.3% falls within this established range, confirming that the engineered feature set is competitive with state-of-the-art approaches. The persistent difficulty of draw prediction is consistent with the findings of Foulley [11], who shows that draw outcomes consistently display the weakest resolution and discrimination across all evaluated models and benchmarks. The natural calibration of Logistic Regression without any post-hoc correction is consistent with the findings of Wheatcroft and Sienkiewicz [6], who demonstrate that careful optimization of a proper score during training can substitute calibration correction. The failure of Isotonic Regression on the 380-match test set directly confirms its known requirement for large calibration datasets as established by Kull et al. [12]. The effectiveness of Temperature Scaling in improving calibration without degrading classification performance is consistent with the findings of Guo et al. [9], who demonstrate this property across a range of modern machine learning models.

D. Objectives and limitations

This paper set out to address two research questions: whether standard machine learning models produce well-calibrated results for football match outcomes, and whether post-hoc calibration methods can improve probabilistic quality without degrading classification accuracy. Both objectives were met. All four models were systematically evaluated across classification and calibration dimensions, and post-hoc calibration with Temperature Scaling successfully improved the probabilistic quality of Random Forest and XGBoost to at or beyond the bookmaker benchmark, while preserving classification performance in both cases.

Several limitations of this work should be acknowledged. The framework is restricted to a single league, Spanish La Liga, and evaluated on a single held-out season of 380 matches, which limits the generalization of the findings across different competitions and time periods. The feature set is limited to pre-match features and does not incorporate in-play data or player-level statistics, which may capture additional predictive signals. A single global Temperature Scaling parameter was applied across all outcome classes, yet the per-class results in Table IV show that calibration improvement was not uniform across home win, draw, and away win outcomes, suggesting that class-specific calibration correction may be more appropriate. Finally, the study does not investigate the relationship between calibration quality and betting profitability, which remains an important direction for future work.

VI. Conclusion

This paper presented a machine learning framework for calibrated forecasting of football match outcomes, with calibration metrics as the final evaluation criterion. Four machine learning models - Logistic Regression, Random Forest, XGBoost, and AutoGluon - were trained on ten seasons of Spanish La Liga data and assessed using a holistic array of classification and calibration diagnostics, benchmarked against betting market-implied probabilities. The main contributions of this work are threefold: a systematic calibration-first evaluation framework applied consistently across four models of varying architectural complexity, a direct benchmarking of model-implied probabilities against professional bookmaker probabilities, and empirical evidence that classification accuracy and probability calibration are independent properties with direct implications for model selection in probabilistic forecasting applications.

On classification performance, all models outperformed the bookmaker baseline with AutoML model AutoGluon achieving the best results in terms of accuracy. Draw prediction remained the central problem of outcome prediction across all models, a finding consistent with the broader literature. The binary reformulation of the task confirmed that the significant prediction error is a direct consequence of the hard predictive nature of the draw class, rather than a limitation of model selection or architecture.

As for calibration, the results demonstrate that achieving a well-calibrated probabilistic forecast is feasible with machine learning models, and post-hoc correction with a single parameter sufficiently improves calibration metrics and achieves the professional bookmaker baseline. Logistic Regression showed well-calibrated results without any corrections, while XGBoost and Random Forest required Temperature Scaling to achieve competitive performance. Notably, XGBoost demonstrated that a model built on bookmaker-derived features can be successfully recalibrated to produce reliable probability estimates. Isotonic Regression consistently worsened the model's probability performance, which is directly connected to the small size of the test dataset.

The most important finding of this work is that classification accuracy and probability quality are independent properties. A model can achieve high accuracy while producing unreliable probabilities, and on the other hand, a well-calibrated model can have relatively modest accuracy. This discrepancy has direct implications: in any context where probability estimates drive financial decisions, such as risk assessment, betting, and probabilistic model comparison, calibration should be evaluated explicitly and should not be inferred from classification performance alone.

The two research questions posed at the outset of this paper were both addressed. Standard machine learning models were shown to produce varying levels of calibration quality, with only Logistic Regression achieving benchmark-level calibration naturally, while post-hoc calibration methods were

shown to improve probabilistic quality of ensemble models without degrading classification accuracy, directly answering both research questions.

The primary limitations of this work include its restriction to a single league and a single held-out season of 380 matches, the use of pre-match features only, and the application of a single global Temperature Scaling parameter across all outcome classes despite non-uniform per-class calibration improvement.

Several directions for future work have been identified. First, a single global temperature value should not be used for every class, as the improvement may not be uniform across all the classes. Then, this framework can be applied to a larger leagues and seasons dataset to properly assess the generalizability of these findings. Finally, if this framework is used for betting strategies or bookmaker financial decisions, the relationship between probability quality and profitability warrants should be properly investigated to identify positive expected returns on investment in practice.

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REFERENCES

[1] F. Rodrigues and A. Pinto, "Prediction of football match results with machine learning," 2022. [Online]. Available: https://www.researchgate.net/publication/363472367

[2] M. A. Rahman, "A deep learning framework for football match prediction," *Discover Applied Sciences*, Jan. 2020. [Online]. Available: https://link.springer.com

[3] B. S. Choi, L. K. Foo, and S.-L. Chua, "Predicting football match outcomes with machine learning approaches," *MENDEL*, vol. 29, no. 2, pp. 229–240, 2023. https://doi.org/10.13164/mendel.2023.2.229

[4] E. Christoffersson, "Beating the odds: Machine learning for football match prediction," 2023. [Online]. Available: https://www.diva-portal.org/smash/record.jsf?pid=diva2:1772002

[5] C. Herbinet, "Using machine learning techniques to predict the outcome of professional football matches," Undergraduate Project Report, Imperial College London, 2018. [Online]. Available: https://www.imperial.ac.uk/media/imperial-college/ faculty-of-engineering/computing/public/distinguished-projects/ 1718-ug-projects/Corentin-Herbinet-Using-Machine-Learning-techniques-to-predict-the-outcome-of-professional-football-matches.pdf

[6] E. Wheatcroft and E. Sienkiewicz, "Calibration and hyperparameter tuning in football forecasting with machine learning," 2023. [Online]. Available: https://www.researchgate.net/publication/352846677

[7] E. Wheatcroft, "Evaluating probabilistic forecasts of football matches: the case against the ranked probability score," *Journal of Quantitative Analysis in Sports*, vol. 17, no. 4, pp. 273–287, Dec. 2021. https://doi.org/10.1515/jqas-2019-0089

[8] M. A. Diniz, R. Izbicki, D. Lopes, and L. E. Salasar, "Comparing probabilistic predictive models applied to football," *Journal of the Operational Research Society*, vol. 70, no. 5, pp. 770–782, May 2019. https://doi.org/10.1080/01605682.2018.1457485

[9] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, Sydney, Australia, 2017, pp. 1321–1330. [Online]. Available: https://arxiv.org/pdf/1706.04599

[10] R. M. Galekwa, J. M. Tshimula, E. G. Tajeuna, and K. Kyandoghere, "A systematic review of machine learning in sports betting: Techniques, challenges, and future directions," arXiv:2410.21484, Oct. 2024. [Online]. Available: https://arxiv.org/abs/2410.21484

[11] J.-L. Foulley, "More on verification of probability forecasts for football outcomes: score decompositions, reliability, and discrimination analyses," arXiv:2106.14345, Jun. 2021. [Online]. Available: https://arxiv.org/abs/2106.14345

[12] M. Kull, T. Silva Filho, and P. Flach, "Beta calibration: A well-founded and easily implemented improvement on logistic calibration for binary classifiers," in *Proc. 20th Int. Conf. Artif. Intell. Stat. (AISTATS)*, Fort Lauderdale, FL, USA, 2017. [Online]. Available: https://proceedings.mlr.press/v54/kull17a.html

[13] Football-Data.co.uk, "Football data and results," [Online]. Available: https://www.football-data.co.uk